CALORIMETRY

ICSE 10

CHAPTER NOTES

Section 11.1: Introduction to Heat

Definition of Heat
Heat is a form of energy that causes a change in temperature or state of a substance. It flows from a region of higher temperature to a region of lower temperature.

Nature of Heat
Heat is a form of kinetic energy associated with the random motion of particles in a substance. The faster the particles move, the higher the temperature.

Units of Heat

- SI unit of heat is Joule (J).
- Another common unit is Calorie (cal), where 1 cal = 4.18 J.

Modes of Heat Transfer

1. **Conduction**: Transfer of heat through solids from particle to particle.
2. **Convection**: Transfer of heat in liquids and gases through the movement of particles.
3. **Radiation**: Transfer of heat in the form of electromagnetic waves without requiring a medium.

Section 11.2: Heat Capacity and Specific Heat Capacity

Heat Capacity (C)
Heat capacity is the amount of heat required to raise the temperature of a body by 1°C.

$$C = \frac{Q}{\Delta T}$$

Where,
Q = heat energy (Joules)
ΔT = change in temperature (°C)

Specific Heat Capacity (c)
The specific heat capacity of a substance is the amount of heat required to raise the temperature of 1 kg of the substance by 1°C.

$$Q = mc\Delta T$$

Where,
Q = heat energy (Joules)
m = mass of the substance (kg)
c = specific heat capacity (J/kg°C)
ΔT = change in temperature (°C)

<u>Difference between Heat Capacity and Specific Heat Capacity</u>

Heat Capacity	Specific Heat Capacity
Depends on the mass of the object	Independent of mass
Unit: J/°C	Unit: J/kg°C
Refers to the whole object	Refers to unit mass of the substance

Section 11.3: Principle of Calorimetry

Principle of Calorimetry
It is based on the law of conservation of energy. In an isolated system, the total heat lost by hot bodies is equal to the total heat gained by cold bodies.

Mathematical Expression

$$m_1c_1(T_1 - T) = m_2c_2(T - T_2)$$

Where,
m_1,m_2 = masses of two bodies
c_1,c_2 = specific heat capacities of two bodies
T_1,T_2 = initial temperatures
T = final equilibrium temperature

Applications of Calorimetry

- Determining specific heat capacities of substances
- Determining latent heat of fusion and vaporization
- Used in thermos flasks to minimize heat loss or gain

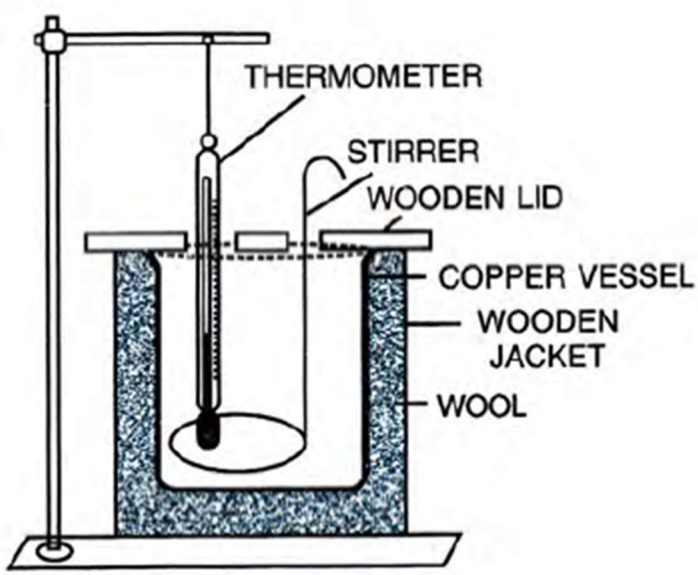

Fig. 11.1 Calorimeter

Section 11.4: Freezing Point and Melting Point

Melting Point
The temperature at which a solid changes into a liquid is called its melting point. For example, the melting point of ice is 0°C.

Freezing Point
The temperature at which a liquid changes into a solid is called its freezing point. For example, the freezing point of water is 0°C.

Effect of Pressure on Melting Point
When pressure increases, the melting point of ice decreases.

Latent Heat of Fusion
The amount of heat required to change 1 kg of a solid into a liquid at constant temperature is called the latent heat of fusion.
Unit: Joules/kg (J/kg)

Section 11.5: Specific Latent Heat

Definition
Specific latent heat is the amount of heat required to change the state of 1 kg of a substance without changing its temperature.

Types of Specific Latent Heat

1. **Latent Heat of Fusion**: It is the heat required to change 1 kg of solid to liquid at a constant temperature.
2. **Latent Heat of Vaporisation**: It is the heat required to change 1 kg of liquid to vapor at a constant temperature.

Formula

$Q = mL$

Where,
Q= heat energy (Joules) , m = mass (kg), L = latent heat (J/kg)

Units : SI unit: Joules per kilogram (J/kg)

Values for Water

- Latent heat of fusion of ice: 334,000 J/kg
- Latent heat of vaporization of water: 2,260,000 J/kg

Section 11.6: Vaporisation

Definition
Vaporization is the process by which a liquid changes into vapor.

Types of Vaporization

1. **Evaporation**: Occurs at the surface of a liquid at all temperatures.
2. **Boiling**: Occurs throughout the liquid at a fixed temperature called the boiling point.

Boiling Point
The temperature at which the vapor pressure of a liquid becomes equal to atmospheric pressure. For water, the boiling point is 100°C under normal atmospheric pressure.

Factors Affecting Boiling Point

1. **Pressure**: Higher pressure increases the boiling point.
2. **Nature of the Liquid**: Different liquids have different boiling points.

Latent Heat of Vaporisation
It is the heat required to convert 1 kg of liquid into vapor at constant temperature.

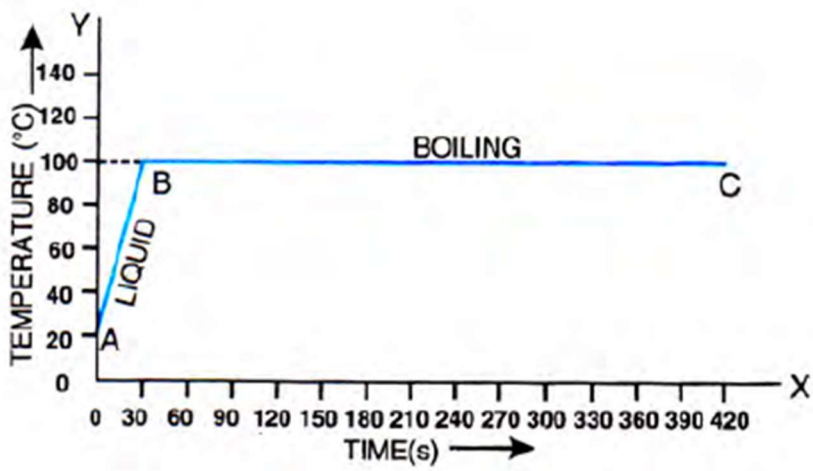

Fig 11.4 Heating curve for water

Section 11.7: Greenhouse Effect

Definition
The greenhouse effect is the trapping of heat in the Earth's atmosphere by greenhouse gases like carbon dioxide, methane, and water vapor.

How It Works

- Solar radiation enters the Earth's atmosphere.
- Earth's surface absorbs the radiation and emits it as infrared (IR) radiation.
- Greenhouse gases trap this IR radiation, preventing it from escaping into space.

Greenhouse Gases

- Carbon dioxide (CO_2)
- Methane (CH_4)
- Nitrous oxide (N_2O)
- Water vapor (H_2O)
- Ozone (O_3)

Causes of the Greenhouse Effect

- Burning of fossil fuels
- Deforestation
- Industrial processes

Effects of the Greenhouse Effect

- **Global Warming**: Rise in the average global temperature.
- **Climate Change**: Extreme weather conditions like floods and droughts.
- **Melting of Glaciers**: Leads to a rise in sea levels.

Ways to Reduce the Greenhouse Effect

- Use of renewable energy sources.
- Reforestation and afforestation.
- Reduction in the use of fossil fuels.